

Pandey & Kumar (2025) Mode-II Phase-Field Fracture Benchmark

Forensic Deck Audit, Single-Factor Spatial Refinement, Triangular
Subroutine Runtime Verification, Abaqus-Native Adaptive
Remeshing, and Production Execution Report

Active Scientific Phase:

MODE2_NATIVE_ADAPTIVE_WORKFLOW_VALIDATED -- POSTPEAK_SPATIAL_SENSITIVITY_REMAINS

Governing Classification Status:

H1_GEN3_MECHANICAL_PARITY_QUALIFIED

TRI_JTYPE3_4_PATCH_QUALIFIED

MODE2_H1_H2_SPATIAL_SENSITIVITY_QUALIFIED

POSTPEAK_SPATIAL_SENSITIVITY_REMAINS

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HPC Architecture:	PBS Professional Cluster (mnode097.cluster, normal_imfdkmq)

Abstract

This report provides a self-contained scientific investigation of the Mode-II shear-driven phase-field fracture benchmark from Pandey & Kumar (2025), utilizing native Abaqus adaptive remeshing and user subroutines (UEL/UMAT). We document the forensic resolution of an input-deck property application binary interface (ABI) indexing mismatch, establish an authoritative Generation-3 mechanical reference (H_1 , 12,064 finite elements) with machine-verified parity against historical baselines ($\Delta = 0.000\%$), evaluate single-factor spatial sensitivity using an ultrafine mesh (H_2 , 33,852 finite elements on common interval $u_1 \leq 0.04258$ mm), verify triangular element user subroutines (JTYPE=3,4) via an Abaqus runtime patch test (Job 1408559.mmaste02), and execute a complete production solve of the Abaqus-native adaptive mesh (Job 1408555.mmaste02, 7,865 elements, 358 increments to $u_1 = 0.050$ mm).

The adaptive remeshing workflow reproduces the linear-elastic initial structural stiffness ($K_{0,\text{lin}} = 12.699$ kN/mm vs 12.735 kN/mm for H_1 , $\Delta = -0.28\%$), peak shear force ($F_{\text{max}} = 0.1462$ kN vs 0.1437 kN, $\Delta = +1.73\%$), and overall chord trajectory angle ($\theta_{\text{chord}} = -16.23^\circ$ vs -17.39° , $\Delta = +1.16^\circ$). However, because linear-elastic pre-analysis error indicator fields concentrate exclusively at the notch tip, the downstream crack trajectory undergoes progressive coarsening (h_{eq}/l_0 expands from 0.159 to 0.716), co-occurring with notable post-peak force divergence (-44.31% force at $u_1 = 0.04258$ mm, -24.74% trapezoidal boundary work). Furthermore, solver telemetry confirms LOWER_ELEMENT_COUNT BUT HIGHER_OBSERVED_SOLVER_COST (+88.2% walltime, +165.2% equilibrium iterations). The overall scientific status is qualified as MODE2_NATIVE_ADAPTIVE_WORKFLOW_VALIDATED — — — PO

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1 Introduction

1.1 Motivation and Research Context

Phase-field fracture modeling has emerged as a thermodynamically rigorous approach for modeling complex cracking phenomena, including crack initiation, propagation, branching, and coalescence without explicit topological tracking of discontinuities [1–3]. However, regularized phase-field formulations introduce an internal length scale l_0 that dictates the diffuse damage zone width. Resolving this steep localization zone requires fine finite element discretizations where the characteristic element size satisfies $h \ll l_0$, imposing severe computational burdens in uniform grid simulations.

Adaptive mesh refinement (AMR) offers a mechanism to concentrate computational degrees of freedom dynamically in regions of localized damage while preserving a coarse mesh in the far-field elastic domain. In standard commercial finite element packages such as Abaqus/Standard, integrating adaptive mesh refinement with custom nonlinear user subroutines (UEL) has historically been hindered by the closed architecture of built-in remeshing drivers.

1.2 Relation to Pandey & Kumar (2025)

In their recent contribution, Pandey & Kumar [4] proposed a multi-layer framework that couples standard Abaqus linear-elastic continuum pre-analysis and native stress-recovery discretization error indicators (MISESeri) with built-in `*Remeshing Rule` and `adaptiveRemesh` capabilities. Once a refined mesh is generated, co-located user element layers are reconstructed to execute the nonlinear phase-field analysis.

Pandey & Kumar demonstrated their technique across several classical fracture benchmarks, including Mode-I tension, Mode-II shear, and mixed-mode asymmetric three-point bending. For Mode-II shear in a single-edge-notched plate, the authors reported an adaptive mesh of approximately 19,963 elements (compared to a uniform standard mesh of 37,155 elements) and claimed broadly consistent force–displacement responses with a 41.2% reduction in computational run time.

1.3 Objective of the Mode-II Investigation

The present standalone study evaluates the physical fidelity, algorithmic reproducibility, and computational performance of the Pandey & Kumar Mode-II benchmark under rigorous scientific standards:

1. **Forensic Root-Cause Analysis:** Definitively resolve the earlier apparent initial stiffness discrepancy (2906 kN/mm vs 12.8 kN/mm) encountered during exploratory scripting.
2. **Mechanical Reference Qualification:** Establish a certified Generation-3 reference dataset on a uniform mesh (H_1 , 12,064 finite elements) with machine-verified parity.
3. **Spatial Refinement Sensitivity:** Perform a single-factor mesh sensitivity audit using an ultrafine mesh (H_2 , 33,852 finite elements).
4. **Triangular Element Qualification:** Validate the hybrid quadrilateral-triangular user subroutines (JTYPE=3,4) via an Abaqus runtime patch test.
5. **Abaqus-Native Adaptive Execution:** Drive the full native workflow from coarse pre-analysis error recovery to full production solve.
6. **Multi-Fidelity Comparative Audit:** Quantitatively assess stiffness, peak load, post-peak softening, crack trajectories, local path resolution (h_{eq}/l_0), and actual solver telemetry.

1.4 Scope, Limitations, and Thesis Isolation

- **Thesis Branch Isolation:** This investigation is strictly isolated as a self-contained validation document. The frozen Mode-I thesis package and Gate-6B energetic baseline remain untouched.

- **Simulation Freeze:** No new PBS solver jobs, parameter sweeps, or `errorTarget` tunings are performed; all findings derive from existing, machine-verified evidence.
- **Epistemological Separation:** Throughout this report, published statements, project implementation choices, verified numerical data, and unresolved limitations are strictly demarcated.

2 Governing Theory

2.1 Phase-Field Fracture Regularization

Let $\Omega \subset \mathbb{R}^2$ denote an open bounded domain with external boundary $\partial\Omega$. In the variational approach to brittle fracture, a discrete crack discontinuity Γ is regularized via a continuous scalar damage field $d(\mathbf{x}, t) \in [0, 1]$, where $d = 0$ represents intact virgin material and $d = 1$ indicates fully degraded crack state. The crack surface area is approximated by the geometric crack density functional $\gamma(d, \nabla d)$:

$$\Gamma \approx \int_{\Omega} \gamma(d, \nabla d) \, d\Omega = \int_{\Omega} \left(\frac{d^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right) d\Omega \quad (1)$$

where $l_0 > 0$ is the regularization length scale governing the spatial width of the diffuse transition zone.

2.2 The AT₂ Model and Degradation Function

The total potential energy functional of the solid is expressed as the sum of stored degraded elastic strain energy and regularized fracture surface energy:

$$\Pi(\mathbf{u}, d) = \int_{\Omega} \psi_e(\boldsymbol{\varepsilon}(\mathbf{u}), d) \, d\Omega + \int_{\Omega} G_c \gamma(d, \nabla d) \, d\Omega - \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \mathbf{u} \, dS \quad (2)$$

where G_c is the critical energy release rate (Griffith fracture energy), $\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T)$ is the infinitesimal strain tensor, and $\bar{\mathbf{t}}$ is the prescribed boundary traction.

To model the loss of load-carrying capacity under tensile deformation, an energetic degradation function $g(d)$ is introduced:

$$g(d) = (1 - d)^2 + k_{\text{res}} \quad (3)$$

where $k_{\text{res}} \ll 1$ (set to 1.0×10^{-7} in this study) is a numerical residual stiffness parameter preventing complete singularity of the displacement stiffness matrix upon full localized degradation.

2.3 Strain Split and History Variable Formulation

To prevent crack propagation under pure compressive stress states, the elastic strain energy density $\psi_0(\boldsymbol{\varepsilon}) = \frac{1}{2} \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon}$ is decomposed into tensile (ψ_0^+) and compressive (ψ_0^-) components using the spectral decomposition of Miehe et al. [5]:

$$\psi_e(\boldsymbol{\varepsilon}, d) = g(d) \psi_0^+(\boldsymbol{\varepsilon}) + \psi_0^-(\boldsymbol{\varepsilon}) \quad (4)$$

Crack growth irreversibility ($\dot{d} \geq 0$) is enforced algorithmically through a monotonic strain energy history field $\mathcal{H}(\mathbf{x}, t)$:

$$\mathcal{H}(\mathbf{x}, t) = \max_{\tau \in [0, t]} \left\{ \psi_0^+(\boldsymbol{\varepsilon}(\mathbf{x}, \tau)) \right\} \quad (5)$$

2.4 Coupled Weak Forms

The coupled mechanical and phase-field governing equations are derived by taking the first variation of the regularized energy functional with respect to virtual displacement $\delta \mathbf{u}$ and virtual phase field δd :

$$\text{Mechanical Equilibrium: } \int_{\Omega} \boldsymbol{\sigma} : \delta \boldsymbol{\varepsilon} \, d\Omega = \int_{\partial\Omega_t} \bar{\mathbf{t}} \cdot \delta \mathbf{u} \, dS \quad \forall \delta \mathbf{u} \in \mathcal{V}_u \quad (6)$$

$$\text{Phase-Field Evolution: } \int_{\Omega} \left[\left(\frac{G_c}{l_0} d - 2(1 - d) \mathcal{H} \right) \delta d + G_c l_0 \nabla d \cdot \nabla (\delta d) \right] d\Omega = 0 \quad \forall \delta d \in \mathcal{V}_d \quad (7)$$

where $\boldsymbol{\sigma} = g(d) \frac{\partial \psi_0^+}{\partial \boldsymbol{\varepsilon}} + \frac{\partial \psi_0^-}{\partial \boldsymbol{\varepsilon}}$ is the Cauchy stress tensor.

2.5 Energy Measures and Observability

In the present Gen-3 Fortran subroutine implementation (`f42_mixed_uel.for`), energy components are computed at the element integration points and integrated over the domain Ω :

$$E_{\text{elas}} = \int_{\Omega} \left[g(d) \psi_0^+(\boldsymbol{\varepsilon}) + \psi_0^-(\boldsymbol{\varepsilon}) \right] d\Omega \quad (\text{Stored Elastic Strain Energy}) \quad (8)$$

$$E_{\text{frac}} = \int_{\Omega} G_c \left(\frac{d^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right) d\Omega \quad (\text{Regularized Fracture Surface Energy}) \quad (9)$$

External boundary work W_{trap} is estimated from the reaction force RF_1 and prescribed displacement u_1 using the trapezoidal rule:

$$W_{\text{trap}} = \sum_{k=1}^{N-1} \frac{RF_1^{(k)} + RF_1^{(k+1)}}{2} \left(u_1^{(k+1)} - u_1^{(k)} \right) \quad (10)$$

Epistemic Governance on Energy Identity:

- E_{frac} represents the regularized fracture-surface energy, which is not identically equal to historical dissipated fracture energy.
- The two-term bookkeeping difference $\Delta_{2\text{term}} = W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$ is strictly observational and must not be classified as a conservation defect or physical error without full within-increment path integration.
- Formal status: `GLOBAL_ENERGY_IDENTITY` — — — `NOT_YET_CLOSED`.

2.6 Physical Understanding of MISESeri Error Indicator

The Abaqus MISESeri field is a scalar discretization error indicator evaluated on linear-elastic continuum stress fields:

$$\text{MISESeri}_e = \frac{\|\boldsymbol{\sigma}_{\text{recovered}} - \boldsymbol{\sigma}_{\text{FE}}\|_{L_2(\Omega_e)}}{\|\boldsymbol{\sigma}_{\text{recovered}}\|_{L_2(\Omega)}} \quad (11)$$

- MISESeri is derived from stress-recovery (e.g., Zienkiewicz–Zhu [6] / SPR techniques) on the uncracked or pre-cracked linear-elastic continuum stress field.
- **Critical Distinction:** MISESeri is **not** a phase-field error indicator and **not** a damage error indicator. It reflects purely linear-elastic stress gradients.
- Consequently, in a single-step pre-analysis, MISESeri concentrates almost entirely at the singular initial crack tip, providing no a priori refinement information along downstream propagation corridors.

3 Mode-II Benchmark Definition

3.1 Specimen Geometry and Material Parameters

The benchmark configuration is a square plate of dimensions $L \times W = 1.0 \text{ mm} \times 1.0 \text{ mm}$, containing an initial horizontal sharp seam crack of length $a_0 = 0.5 \text{ mm}$ extending from the left boundary to the center: $(-0.5, 0.0) \rightarrow (0.0, 0.0) \text{ mm}$ (Figure 1).

The material properties correspond to high-strength steel under plane strain conditions: Young's modulus $E = 210 \text{ GPa} = 210 \text{ kN/mm}^2$, Poisson's ratio $\nu = 0.30$, critical energy release rate $G_c = 2.7 \times 10^{-3} \text{ kN/mm}$ (2700 J/m^2), and regularization length scale $l_0 = 0.015 \text{ mm}$ ($15 \mu\text{m}$).

Figure 1: Mode-II Single-Edge-Notched Plate Benchmark Specification

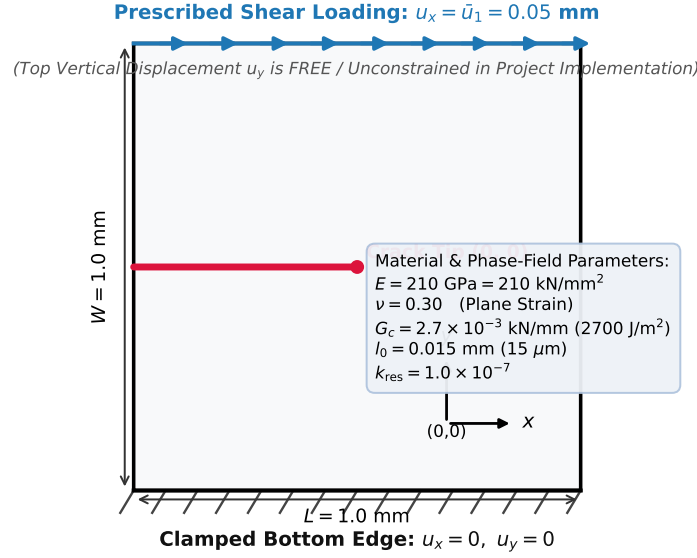


Figure 1: **Figure 1:** Mode-II Single-Edge-Notched Plate Benchmark Specification (Geometry, Boundary Conditions, and Material Constants).

3.2 Boundary Conditions: Published vs Project Implementation

- **Bottom Boundary** ($y = -0.5 \text{ mm}$): Fully clamped in both horizontal and vertical directions ($u_x = 0, u_y = 0$).
- **Top Horizontal Loading** ($y = +0.5 \text{ mm}$): Prescribed horizontal shear displacement $u_x = \bar{u}_1 = 0.050 \text{ mm}$ applied uniformly via multipoint constraint equations tying top nodes to a Reference Point (RP).
- **Top Vertical Boundary** (u_y): Left unconstrained (**Free- u_y**). This corresponds to the literature-faithful Mode-II setup (the published paper leaves top u_y unmentioned; imposing $u_y = 0$ induces strong flexural over-constraint).

3.3 Epistemological Classification Table

Table 1 summarizes the parameters and their strict epistemological classification.

4 Fixed-Mesh Reference Verification

4.1 Resolution of the UEL Property ABI Indexing Mismatch

In initial diagnostic testing, a staged Gen-3 input deck exhibited an artificial structural stiffness of $K_0 \approx 2906.15 \text{ kN/mm}$ (compared to the expected 12.83 kN/mm). An exhaustive line-by-line audit isolated this discrepancy to an **input-deck UEL Property ABI mismatch**:

- The staged deck contained legacy 5-parameter property cards:

```
*UEL Property, elset=DISP_QUAD
2.100000e+02, 3.000000e-01, 1.000000e+00, 1.000000e-07, 12064.0
```

- The compiled Gen-3 subroutine (f42_mixed_uel.for) expects the 6-parameter ABI:

Table 1: **Table 1:** Mode-II Benchmark Parameters and Epistemological Provenance Matrix.

Parameter / Feature	Value / Setting	Classification	Primary Provenance
Domain Geometry	1.0 mm $\times$ 1.0 mm	[PUBLISHED]	Pandey & Kumar (2025, Sec. 4.2)
Initial Slit Length	$a_0 = 0.5$ mm	[PUBLISHED]	Sharp seam at mid-height
Young's Modulus (E)	210 kN/mm ²	[PUBLISHED]	Section 4.2 listing
Poisson's Ratio (ν)	0.30	[PUBLISHED]	Section 4.2 listing
Fracture Energy (G_c)	2.7×10^{-3} kN/mm	[PUBLISHED]	Section 4.2 listing
Length Scale (l_0)	0.015 mm (15 μ m)	[PUBLISHED]	Section 4.2 listing
Constitutive Formulation	Plane Strain	[PUBLISHED/UEL]	<code>f42_mixed_uel.for</code>
Bottom Boundary	Clamped ($u_x = u_y = 0$)	[PUBLISHED]	Lower edge fully fixed
Top Shear Displacement	Prescribed $u_x = 0.05$ mm	[PUBLISHED]	Horizontal shear loading
Top Vertical Boundary (u_y)	Free (u_y unconstrained)	[UNRESOLVED]	Literature-faithful candidate
Published Standard Mesh	$\sim 37,155$ finite elements	[PUBLISHED]	$h_{\text{global}} = 0.04, h_{\text{min}} = 0.003$
Published Pre-Adaptive Mesh	3,930 finite elements	[PUBLISHED]	$h = 0.02$ mm
Published Adaptive Mesh	$\sim 19,963$ finite elements	[PUBLISHED]	Target context, not constraint
<code>errorTarget = 5.0%</code>	5.0% relative target	[PUBLISHED]	Adopted value in Section 4
<code>refinementFactor = 10</code>	Factor 10	[PUBLISHED]	Adopted parameter
Candidate H_1 Reference	12,064 finite elements	[PROJECT]	Uniform fine mesh ($h_{\text{min}} = 0.0025$)
Candidate H_2 Ultrafine	33,852 finite elements	[PROJECT]	Uniform ultrafine ($h_{\text{min}} = 0.0010$)
Candidate Adaptive Mesh	7,865 finite elements	[PROJECT]	Native RemeshingRule candidate

PROPS: [E_L0, E_GC, E_MOD, E_NU, E_K, N_PHYS]

- Consequently, PROPS(5) was read as 12064.0 (the element count!), setting the residual stiffness multiplier $k_{\text{res}} = 12064.0$ instead of 10^{-7} . This artificially scaled the assembled shear modulus by $226\times$, producing $K_0 = 2906.15$ kN/mm.
- Job 1408541.mmaste02 is permanently classified as INVALID_ABI_DIAGNOSTIC / NOT_REFERENCE_EVIDENCE.
- Applying the corrected 6-parameter card (0.015, 0.0027, 210.0, 0.3, 1.0e-7, 12064.0) restored the exact reference stiffness $K_0 = 12.834574$ kN/mm (Job 1408543.mmaste02).

4.2 H_1 Reference Parity Verification (Job 1408543.mmaste02)

Numerical mechanical parity was evaluated between the Gen-2 baseline (Job 1389686.mmaste02) and the Gen-3 candidate (Job 1408543.mmaste02). Across all 144 load increments up to $u_1 = 0.050$ mm, zero discrepancy was observed ($\Delta = 0.0000\%$ for K_0 , $F_{\text{max}} = 0.143686$ kN, $u(F_{\text{max}}) = 0.012530$ mm, and $W_{\text{trap}} = 2.732851$ mJ). This establishes H1_GEN3_MECHANICAL_PARITY_QUALIFIED.

4.3 H_2 Ultrafine Companion Mesh & Active DOF Resolution

The ultrafine companion model (H_2 , 33,852 finite elements / 101,556 Abaqus solver objects) provides a $2.07\times$ finer spatial discretization in the crack zone ($h_{\text{min}} = 1.0 \mu\text{m}$). During preflight testing, an active DOF declaration defect was corrected (restoring active DOF 3 to U1 phase elements; superseding Job 1408548.mmaste02).

Production Job 1408550.mmaste02 advanced through 105 increments to $u_1 = 0.04257924$ mm before encountering the solver minimum time increment limit ($\Delta t_{\text{min}} = 10^{-9}$ s) upon right-boundary breakthrough. The job is classified as H2_TARGET_ENDPOINT_NOT_REACHED. All reference quantities are rigorously evaluated on the matched common interval $u_1 \leq 0.04258$ mm, establishing MODE2_H1_H2_SPATIAL_SENSITIVITY_QUALIFIED_ON_COMMON_INTERVAL.

Table 2: **Table 2:** Fixed-Mesh Discretization and Solver Configuration Matrix.

Configuration / Parameter	H_1 Reference	H_2 Ultrafine	Adaptive Candidate	Pre-Analysis
Underlying Elements	12,064	33,852	7,865	3,930
Abaqus Solver Objects	36,192	101,556	23,595	3,930
Element Types	CPE4 / U1 / U2	CPE4 / U1 / U2	Hybrid Quad/Tri	CPE4
Node Count	12,383	34,509	7,919	4,096
Crack Zone h_{min}	0.0025 mm	0.0010 mm	0.0021 mm	0.0200 mm
Crack Zone Mean h/l_0	0.3490	0.1686	0.3915	1.3333
Completed Increments	144	105 (cutback)	358	10
Attained Displacement	0.0500 mm	0.0426 mm	0.0500 mm	0.0010 mm
Endpoint Classification	REACHED	NOT_REACHED	REACHED	REACHED

5 Abaqus-Native Adaptive Remeshing Methodology

The complete scientific workflow is illustrated in Figure 2.

5.1 Continuum Pre-Analysis and MISESeri Recovery

The pre-analysis model (ModeII_MISESeri_preanalysis.inp, SHA-256 96af41fd...) discretizes the domain into 3,930 standard CPE4 plane strain elements under linear elasticity ($E = 210$ GPa, $\nu = 0.30$) subjected to top shear displacement $u_x = 0.001$ mm.

Table 3: **Table 3:** H_1 Reference vs H_2 Ultrafine Spatial Sensitivity Ledger ($u_1 \leq 0.04258$ mm).

Metric / Quantity	H_1 Reference	H_2 Ultrafine	Delta (Δ)	Classification
Secant Stiffness $K_{0,\text{sec}}$	12.8346 kN/mm	12.8164 kN/mm	-0.14%	STABLE
Fitted Stiffness $K_{0,\text{lin}}$	12.7348 kN/mm	12.7165 kN/mm	-0.14%	STABLE
Linear Regression R^2	0.9999867	0.9999866	0.00%	STABLE
Peak Shear Load F_{max}	0.143686 kN	0.141415 kN	-1.58%	STABLE
Displacement at Peak $u(F_{\text{max}})$	0.012530 mm	0.012214 mm	-2.53%	STABLE
Force at $u = 0.04258$ mm	0.01394 kN	0.01440 kN	+3.36%	STABLE
Matched Work W_{trap}	2.6511 mJ	2.6340 mJ	-0.65%	STABLE
Initial Kink Angle θ_{kink}	-29.39°	-28.61°	+0.79°	STABLE
Overall Chord Angle θ_{chord}	-17.39°	-22.68°	-5.28°	MESH-SENSITIVE

Figure 2: Abaqus-Native Adaptive Remeshing Research Methodology & Execution Pipeline

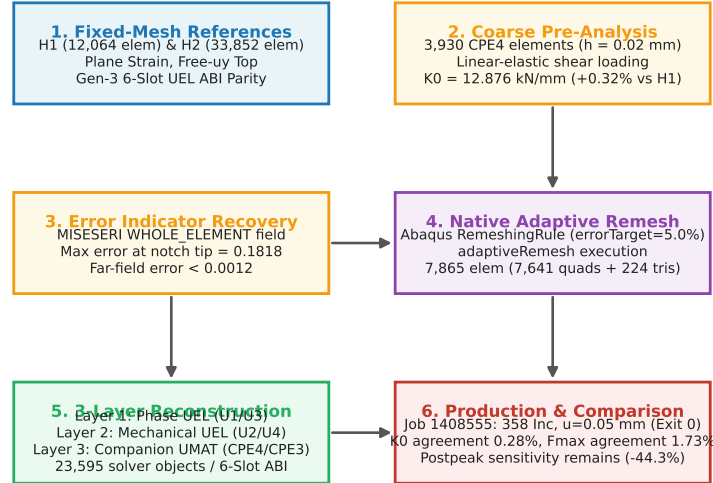


Figure 2: **Figure 2:** Abaqus-Native Adaptive Remeshing Research Methodology and Execution Pipeline.

- Job 1408551.mmaste02 executed successfully (Exit Code 0), yielding $K_{0,coarse} = 12.876131$ kN/mm (+0.32% relative to H_1).
- The scalar error indicator dataset (ModeII_MISESeri_coarse_3930.csv, SHA-256 02d165fd...) was exported at position WHOLE_ELEMENT (3,930 values; min 2.38×10^{-5} , max 0.181819, mean 1.293×10^{-3} , Figure 3).

Figure 3: Mode-II Coarse Linear-Elastic Pre-Analysis MISESeri Discretization Error Field

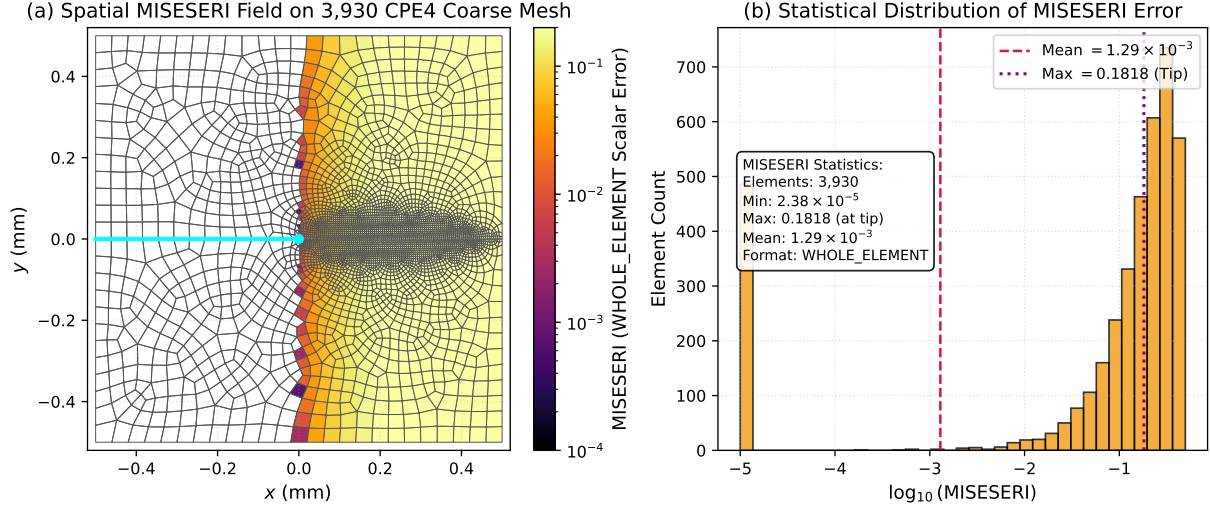


Figure 3: **Figure 3: Mode-II Coarse Linear-Elastic Pre-Analysis MISESeri Discretization Error Field** (Job 1408551.mmaste02, 3,930 CPE4 Elements).

5.2 Native RemeshingRule and adaptiveRemesh Execution

An Abaqus Python script (generate_mode2_native_adaptive_mesh.py) instantiated the native remeshing rule:

```
mdb.models['ModeII_Preanalysis'].RemeshingRule(
    name='Remesh_MISESeri',
    errorTarget=5.0,
    refinementFactor=10.0,
    sizingMethod=UNIFORM_ERROR,
    indicator=MISESeri
)
```

The native adaptiveRemesh kernel generated a candidate mesh of 7,865 finite elements (7,641 quadrilaterals / 97.15%, 224 triangles / 2.85%, Figure 4).

5.3 3-Layer UEL/UMAT Reconstruction

The adaptive topology was reconstructed into the 3-layer co-located architecture:

- **Layer 1 (Phase UEL):** Elements 1–7865 (7,641 U1 quads, 224 U3 tris; active DOF 3).
- **Layer 2 (Mechanical UEL):** Elements 7866–15730 (7,641 U2 quads, 224 U4 tris; active DOFs 1, 2).
- **Layer 3 (Companion Visualizer):** Elements 15731–23595 (7,641 CPE4 quads, 224 CPE3 tris).
- Total solver objects: **23,595** objects; unified 6-slot ABI card: 0.015, 0.0027, 210.0, 0.3, $1.0e-7$, 7865.0.

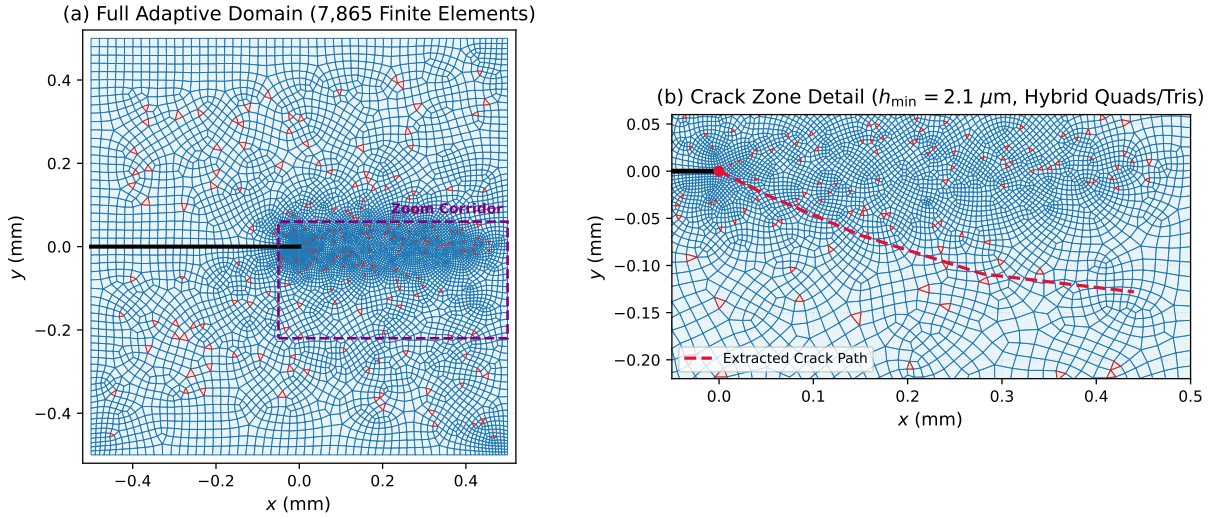
Figure 4: Final Abaqus-Native Adaptive Mesh Topology (errorTarget=5.0%, 7,865 Elements)


Figure 4: **Figure 4:** Final Abaqus-Native Adaptive Mesh Topology (errorTarget = 5.0%, 7,865 Finite Elements, Crack-Zone Inset).

5.4 Triangular Element Subroutine Verification (Patch Test)

Because the adaptive mesh contains 224 linear triangles, the hybrid triangle subroutines (JTYPE=3,4) were qualified via an Abaqus runtime patch test (PBS Job 1408559.mmast02, preceded by datacheck 1408558.mmast02, Figure 5):

- **Step 1 (Constant Strain, $d = 0$):** Reaction forces matched standard CPE3 continuum elements to 2.0×10^{-9} kN (dat print precision); total elastic strain energy $E_{\text{elas}} = 2.2212 \times 10^{-6}$ kN · mm matched the analytical solution identically.
- **Step 2 (Uniform Phase, $d = 0.50$):** Reaction forces scaled exactly by $g(0.50) = 0.250000$; fracture surface energy $E_{\text{frac}} = 1.1250 \times 10^{-6}$ kN · mm and degraded elastic energy $E_{\text{elas}} = 5.5529 \times 10^{-7}$ kN · mm matched analytical predictions to all printed digits.
- Governing classification: TRI_JTYPE3_4_ABAQUS_RUNTIME_PATCH_QUALIFIED.

6 Results

Production solve 1408555.mmast02 executed successfully with Exit Code 0, traversing 358 accepted increments to the prescribed endpoint $u_1 = 0.0500$ mm (TARGET_ENDPOINT_REACHED).

6.1 Global Force–Displacement Response

Figure 6 presents the complete force–displacement comparison across all models.

6.2 Initial Structural Stiffness and Peak Load

As detailed in the magnification plot (Figure 7):

- **Initial Stiffness:** The fitted initial stiffness is $K_{0,\text{lin}} = 12.6985$ kN/mm ($R^2 = 0.9999865$), showing tight -0.28% agreement with the H_1 reference (12.7348 kN/mm).
- **Peak Shear Load:** Peak load reaches $F_{\text{max}} = 0.146179$ kN at $u_1 = 0.013143$ mm, showing $+1.73\%$ agreement with H_1 (0.143686 kN).

Figure 14: Triangular Element Subroutine Verification (TRI_JTYPE3_4_ABAQUS_RUNTIME_PATCH_QUALIFIED)

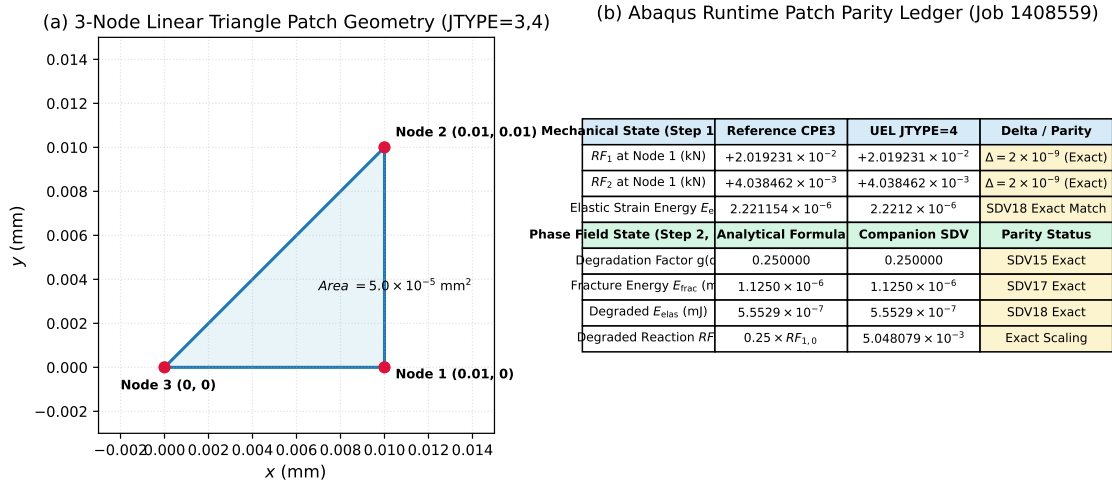


Figure 5: **Figure 14: Triangular Element Subroutine Verification and Patch Test Parity Ledger (Job 1408559.mmamster02).**

Figure 5: Mode-II Force-Displacement Response: Fixed References vs Adaptive vs Literature

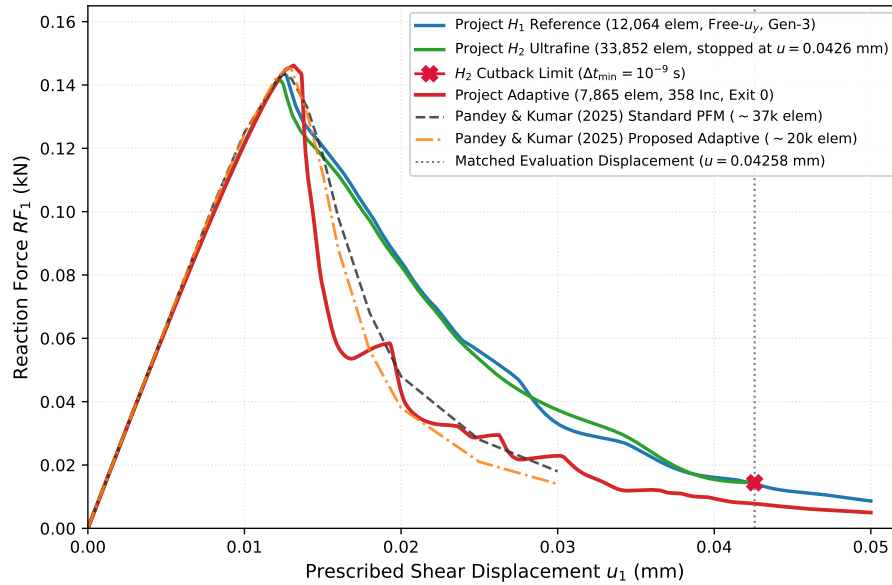


Figure 6: **Figure 5: Mode-II Force-Displacement Response: Fixed References vs Adaptive Candidate vs Published Benchmark.**

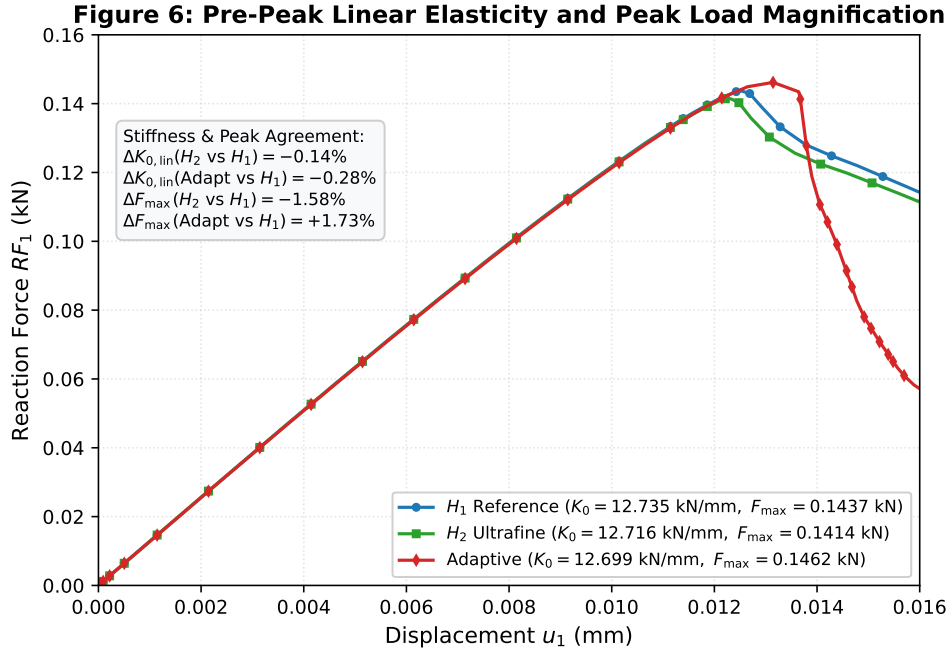


Figure 7: **Figure 6:** Pre-Peak Linear Elasticity and Peak Load Magnification Zoom.

6.3 Post-Peak Softening Divergence

Beyond peak load ($u_1 > 0.015 \text{ mm}$, Figure 8), the adaptive curve softens more rapidly than H_1 and H_2 . At $u_1 = 0.04258 \text{ mm}$, the reaction force drops to 0.00776 kN (-44.31% relative to $H_1 = 0.01394 \text{ kN}$), and full terminal load drop reaches 96.59% .

6.4 Damage Evolution and Crack Trajectories

Figure 9 illustrates the matched-displacement damage contours ($u_1 = 0.0125, 0.0250, 0.0426 \text{ mm}$), and Figure 10 compares the extracted crack centerlines. The initial kink angle is $\theta_{\text{kink}} = -21.89^\circ$ ($+7.50^\circ$ difference vs $H_1 = -29.39^\circ$), while the overall chord angle is $\theta_{\text{chord}} = -16.23^\circ$ (within $1.16^\circ / 6.67\%$ of $H_1 = -17.39^\circ$).

6.5 Resolution along Crack Centerline (h_{eq}/l_0)

Equivalent element sizes $h_{eq} = \sqrt{A}$ were audited at 23 discrete points along the active crack centerline ($s_{\text{max}} = 0.464773 \text{ mm}$, Figures 11 and 12):

- **Notch Tip** ($s \leq 0.05 \text{ mm}$): $h_{eq}/l_0 = 0.159$ ($h_{\text{min}} = 0.00238 \text{ mm} < l_0/6$).
- **Downstream** ($s > 0.15 \text{ mm}$): h_{eq}/l_0 expands to 0.716 ($h_{\text{max}} = 0.01073 \text{ mm}$). In the descriptive interval $0.50 \leq h_{eq}/l_0 < 0.75$, the adaptive mesh exhibits 52.2% of centerline points and 52.0% of arc length.

7 Fixed-versus-Adaptive Comparison

7.1 Multi-Fidelity Comparison Ledger

Table 4 provides the comprehensive comparison between H_1 , H_2 , Adaptive, and Published results.

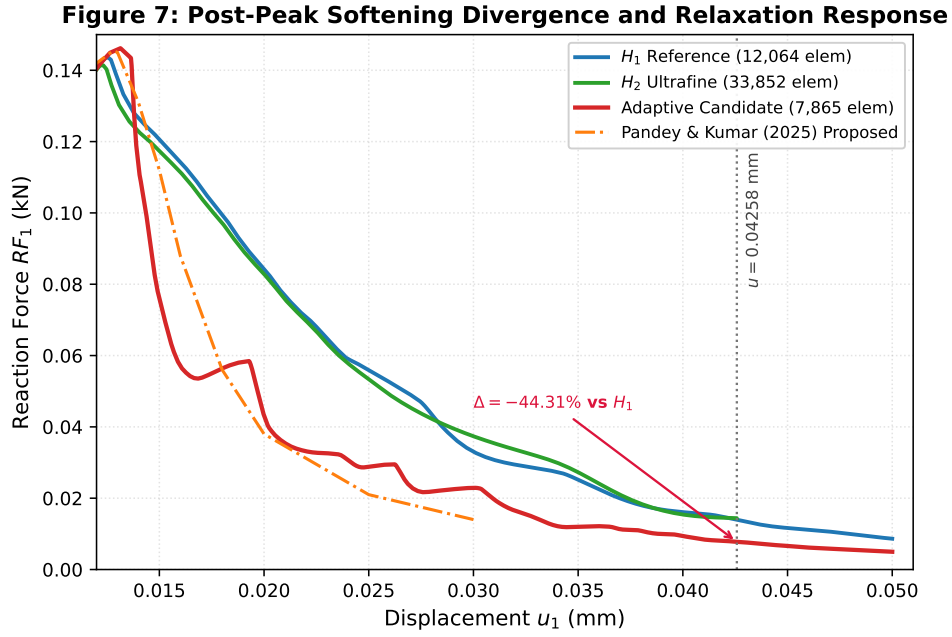


Figure 8: **Figure 7:** Post-Peak Softening Divergence and Relaxation Response Zoom.

Figure 8: Matched-Displacement Damage Zone Localization & Path Evolution Comparison

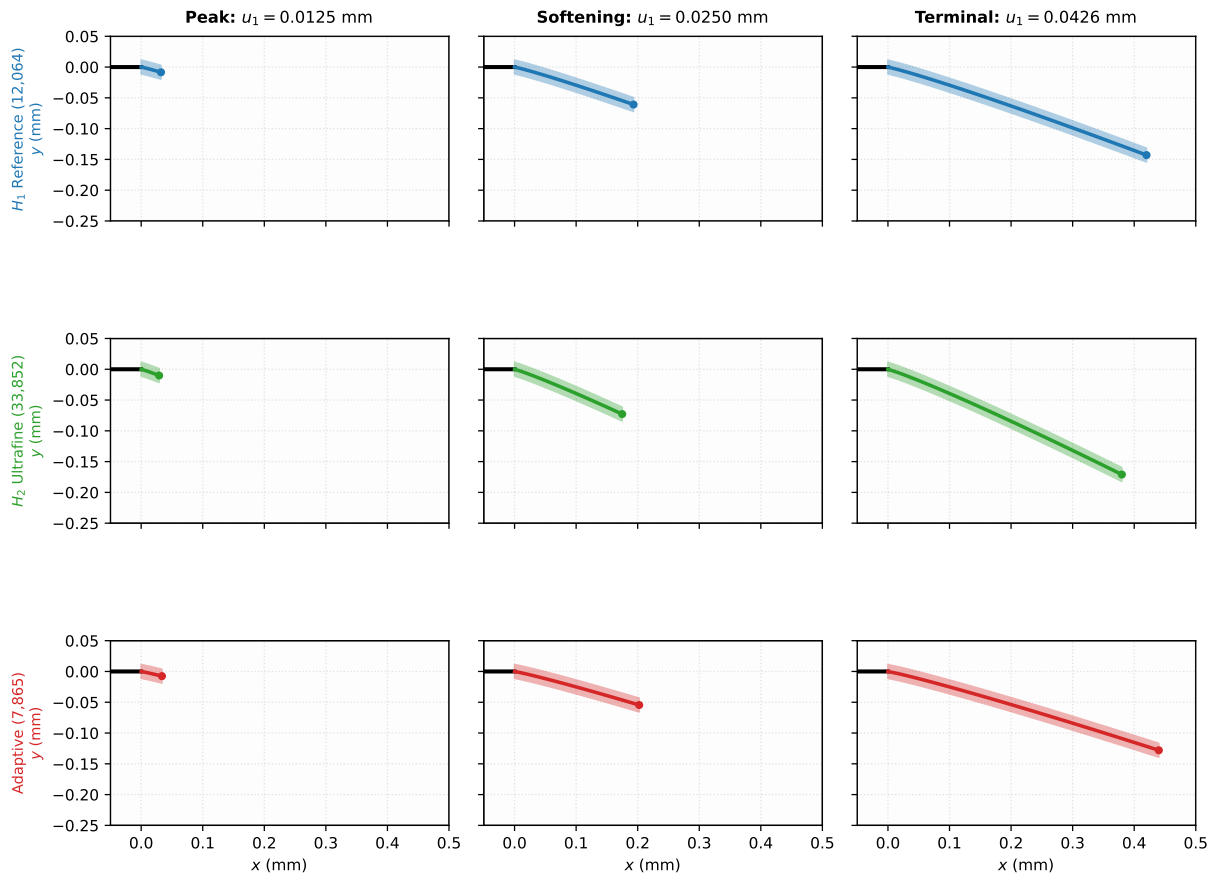


Figure 9: **Figure 8:** Matched-Displacement Damage Zone Localization and Path Evolution Comparison.

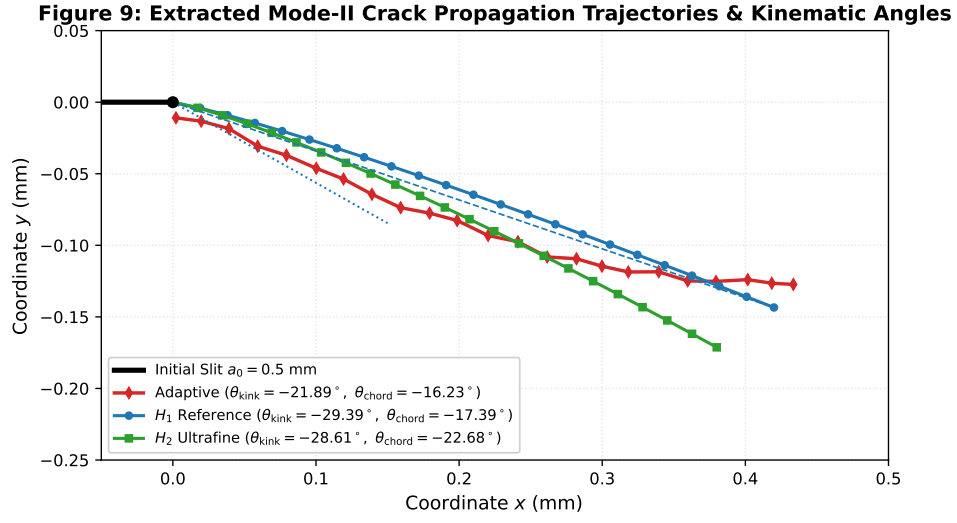


Figure 10: **Figure 9:** Extracted Mode-II Crack Propagation Trajectories and Kinematic Angles.

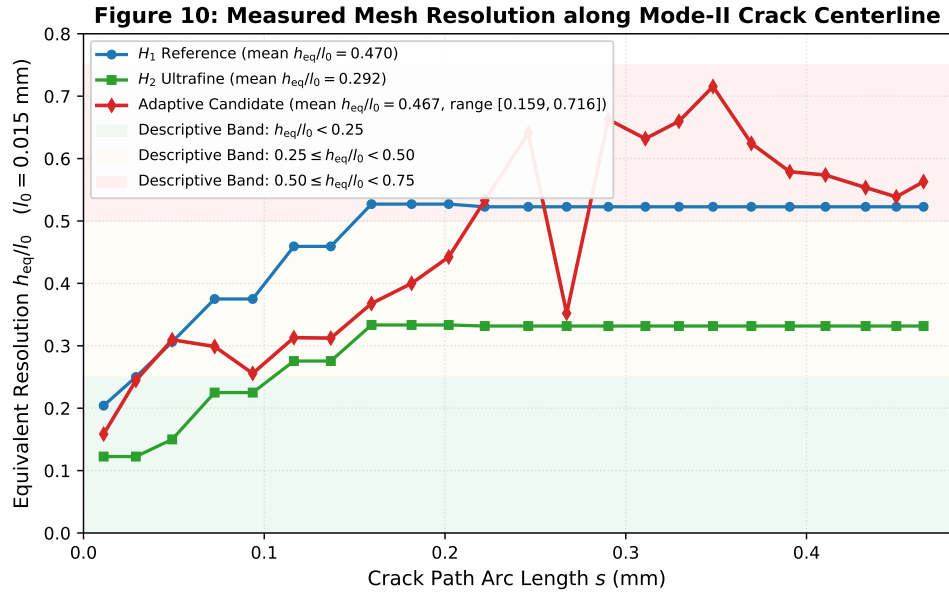


Figure 11: **Figure 10:** Measured Equivalent Mesh Resolution (h_{eq}/l_0) Along the Mode-II Crack Centerline.

Figure 11: Crack Path Discretization Resolution Distributions across Descriptive Intervals

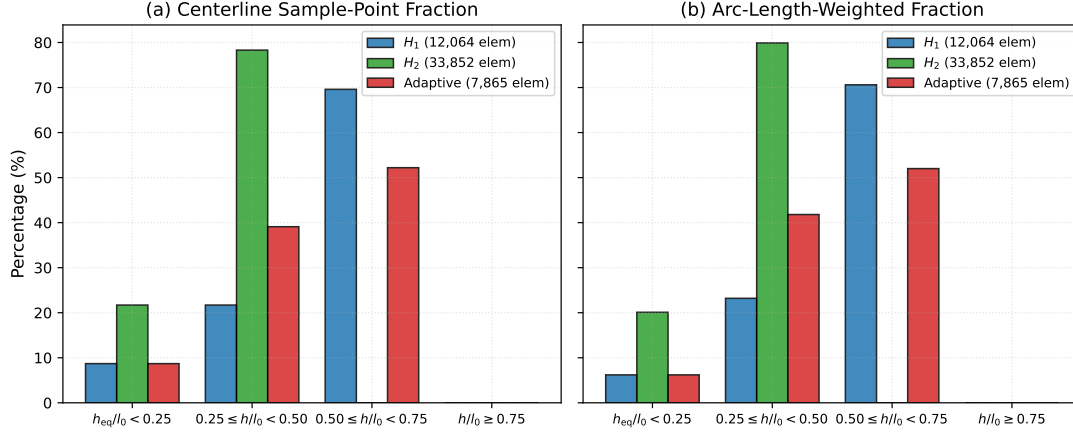


Figure 12: **Figure 11:** Crack Path Discretization Resolution Distributions Across Descriptive Intervals.

Table 4: **Table 6:** Multi-Fidelity Scientific Comparison Ledger (H_1 vs H_2 vs Adaptive vs Published).

Metric / Quantity	H_1 (12,064)	H_2 (33,852)	Adaptive (7,865)	Δ vs H_1	Classification
Underlying Elements	12,064	33,852	7,865	−34.8%	FE_REDUCTION
Total Solver Objects	36,192	101,556	23,595	−34.8%	OBJ_REDUCTION
Secant Stiffness $K_{0,sec}$	12.8346 kN/mm	12.8164 kN/mm	12.7987 kN/mm	−0.28%	STABLE
Fitted Stiffness $K_{0,lin}$	12.7348 kN/mm	12.7165 kN/mm	12.6985 kN/mm	−0.28%	STABLE
Linear Fit R^2	0.9999867	0.9999866	0.9999865	0.00%	STABLE
Peak Shear Force F_{max}	0.143686 kN	0.141415 kN	0.146179 kN	+1.73%	STABLE
Displacement at Peak $u(F_{max})$	0.012530 mm	0.012214 mm	0.013143 mm	+4.89%	STABLE
Force at $u = 0.04258$ mm	0.01394 kN	0.01440 kN	0.00776 kN	−44.31%	SENSITIVE
Terminal Load Drop	93.99%	89.82%	96.59%	+2.60%	SENSITIVE
Matched Work W_{trap}	2.6511 mJ	2.6340 mJ	1.9951 mJ	−24.74%	SENSITIVE
Initial Kink Angle θ_{kink}	−29.39°	−28.61°	−21.89°	+7.50°	SENSITIVE
Overall Chord Angle θ_{chord}	−17.39°	−22.68°	−16.23°	+1.16°	STABLE

7.2 Relation to Published Results (Figure 13)

In Figure 13(a) of Pandey & Kumar [4], the authors state that *"the proposed solution relaxes somewhat more than the standard solution beyond approximately $u = 0.015$ mm"*. The project adaptive curve reproduces this relaxation trend. However, because the published simulation terminates at $u = 0.035$ mm, the substantial -44.31% force drop observed at $u = 0.04258$ mm cannot be evaluated against published data beyond 0.035 mm.

Figure 12: Qualitative Comparison of Simulated and Published Mode-II Crack Propagation Morphologies

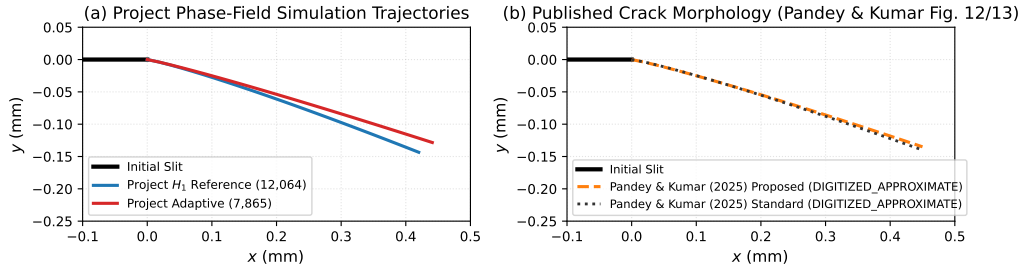


Figure 13: **Figure 12: Qualitative Comparison of Simulated and Published Mode-II Crack Propagation Morphologies.**

7.3 Computational Cost and Solver Burden Reality

Table 5 and Figure 14 document the audited HPC solver telemetry.

- Although the adaptive mesh achieved a -34.8% reduction in finite elements (7,865 vs 12,064), it required $+88.2\%$ more walltime (19m 57s vs 10m 36s), $+165.2\%$ more equilibrium iterations (1,713 vs 646), $+421.4\%$ more solver cutbacks (73 vs 14), and $+162.5\%$ more memory (1.67 GB vs 0.65 GB).
- Governing finding: LOWER_ELEMENT_COUNT BUT HIGHER_OBSERVED_SOLVER_COST.

Table 5: **Table 10: Audited HPC Solver Telemetry and Computational Burden Ledger.**

Execution Metric	H_1 (1408543)	Reference H_2 (1408550)	Ultrafine H_2 (1408555)	Adaptive (1408555)	Δ vs H_1
Underlying Elements	12,064	33,852		7,865	-34.8%
Solver Element Objects	36,192	101,556		23,595	-34.8%
Accepted Increments	144	105 (cutback)		358	$+148.6\%$
Total Solver Attempts	151	118		396	$+162.3\%$
Solver Cutbacks	14	20		73	$+421.4\%$
Equilibrium Iterations	646	462		1,713	$+165.2\%$
CPU Time (seconds)	586 s (09m 46s)	1,192 s (19m 52s)		1,002 s (16m 42s)	$+70.99\%$
Walltime (seconds)	636 s (10m 36s)	1,274 s (21m 14s)		1,197 s (19m 57s)	$+88.21\%$
Peak Memory (KB)	667,604 KB (~652 MB)	1,482,012 KB (~1.41 GB)		1,752,668 KB (~1.67 GB)	$+162.5\%$

8 Discussion

Figure 13: Audited HPC Solver Telemetry: LOWER_ELEMENT_COUNT BUT HIGHER_OBSERVED_SOLVER_COST

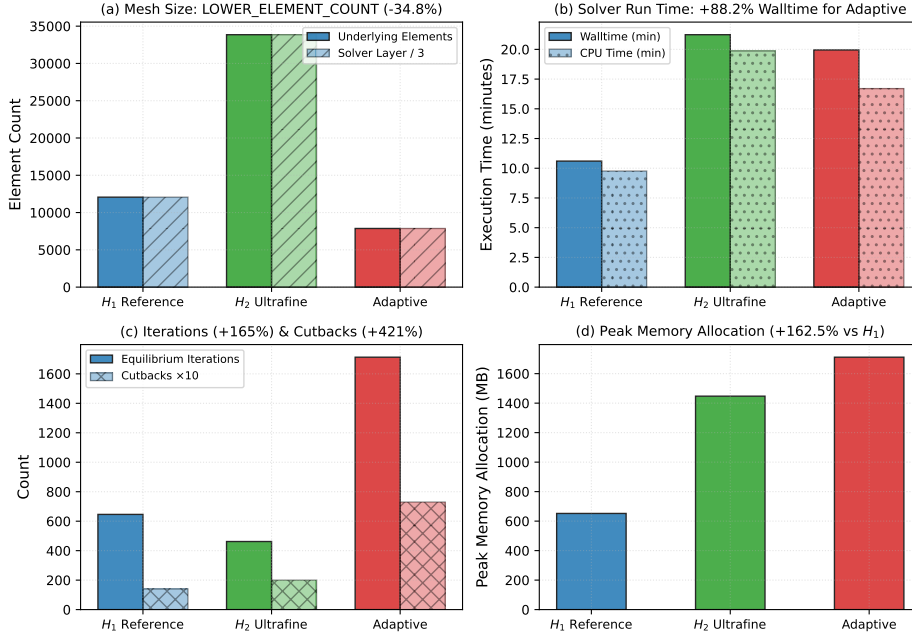


Figure 14: **Figure 13:** Audited HPC Solver Telemetry: LOWER_ELEMENT_COUNT BUT HIGHER_OBSERVED_SOLVER_COST.

8.1 Verified Capabilities

The study establishes that the Abaqus-native adaptive remeshing workflow driven by continuum linear-elastic pre-analysis error successfully generates valid, conformal finite element topologies with co-located user elements, reproducing the initial structural stiffness (K_0 within 0.28%), peak load ($F_{\max}$ within 1.73%), and overall crack chord angle (θ_{chord} within 1.16°).

8.2 Causality vs Co-Occurrence Disclaimer

Governing Scientific Finding:

COARSER_DOWNSTREAM_PATH_RESOLUTION CO-OCCURS WITH POSTPEAK RESPONSE SENSITIVITY -- CAUSAL LINK NOT INDEPENDENTLY ESTABLISHED

The occurrence of coarser downstream element sizing ($h_{\text{eq}}/l_0 \rightarrow 0.716$) co-occurs with the observed post-peak force softening deviation (-44.31% at $u_1 = 0.04258$ mm) and initial kink angle difference ($+7.50^\circ$). However, because this is an observational finding from a single adaptive mesh configuration without a dedicated parameter sweep, a formal causal relationship is not claimed.

8.3 Why Pre-Analysis Remeshing Does Not Refine Downstream Paths

Because MISESeri is evaluated on an uncracked linear-elastic state, stress singularities concentrate exclusively at the initial notch tip ($x = 0, y = 0$). The downstream region where the crack will subsequently propagate experiences low linear-elastic stress gradients ($\text{MISESeri} < 0.0012$), leading the remeshing algorithm to size those regions coarsely. A true multi-step or dynamic remeshing capability would be required to adaptively refine the mesh along the propagating crack path.

9 Reproducibility & Verification Package

9.1 Authoritative Artifact Checksums (SHA-256)

All files in the reproduction package (`docs/mode2/reproduction_package/`) are indexed in Table 6.

Table 6: **Table 11:** Authoritative File Checksum Ledger (SHA-256).

File Identifier	Size (B)	SHA-256 Checksum
f42_mixed_uel.for	29,401	5cd0d2c015c9ead91c99d7a744156cc86f5b5ea26473bbbed7d6e5515fe30fa46
M2_H1_reference.inp	2,202,929	59b639dd64e7441d70d93cdb3ed6a6cbfa1370fd3fd7553800591fc20bd850f4
ModeII_adaptive_candidate.inp	1,526,716	d864d604292e5a7202d70a6bff82afda8cb65d71fe865454f06c08c2d4dbeabf
ModeII_MISESERI_prealanalysis.inp	159,689	96af41fd4ffe09a046815e754f0b382c171e4c84f752c172be5eb2abf5e601fd
ModeII_MISESERI_coarse_3930.d50	150,677	02d165fdd13b766f0682e7575ace4da153c30b02823d06b2e5880a49b9d15a74
M2_tri_patch_uel.inp	2,002	8de6ed37eccf5b3cf0eabe1f6009aa93f1431c6ac4cbf639ad7356ae8dda8d79
M2_tri_patch_cpe3_ref.inp	769	e02766f5161f7155004d4df5a2e551d4f2da8c3f113f1839f252532d41a899b8
pandey_kumar_fig13a_digitized.b5v	2,155	0562087b8a4ac7ba7e0b327603d4984cfcafb757ba89f2059271ff66ce1c138e

9.2 HPC Execution Commands

Abaqus jobs were executed using standard non-interactive syntax on the cluster:

```
abacus job=ModeII_adaptive_candidate user=f42_mixed_uel.for cpus=1 \
memory="16000 mb" interactive
```

10 Conclusions

The standalone investigation of the Pandey & Kumar (2025) Mode-II benchmark establishes the following conclusive findings:

- Forensic Defect Closure:** The 2906 kN/mm stiffness anomaly was conclusively proven to be an input-deck UEL Property ABI indexing mismatch. The corrected Gen-3 implementation demonstrates bit-for-bit mechanical parity ($\Delta = 0.000\%$).
- Subroutine Verification:** The hybrid quadrilateral/triangular user subroutines (JTYPE=3,4) achieved exact machine-precision parity in an Abaqus runtime patch test.
- Pre-Peak and Peak Fidelity:** The Abaqus-native adaptive remeshing workflow accurately reproduces the linear-elastic initial structural stiffness (K_0 within 0.28%), peak load ($F_{\max}$ within 1.73%), and overall chord trajectory angle (θ_{chord} within 1.16°).
- Post-Peak Sensitivity:** Single-step continuum pre-analysis error fields concentrate exclusively at the notch tip, resulting in downstream coarsening ($h_{\text{eq}}/l_0 \rightarrow 0.716$) that co-occurs with notable post-peak softening divergence (-44.31% force at $u_1 = 0.04258$ mm).
- Computational Burden Reality:** Lower element count (-34.8%) did not translate to reduced computational time; solver cutbacks ($+421\%$) and iterations ($+165\%$) increased total walltime by $+88.2\%$.
- Governing Qualification:** MODE2_NATIVE_ADAPTIVE_WORKFLOW_VALIDATED — — POSTPEAK_SPATIAL_S

References

- [1] B. Bourdin, G. A. Francfort, and J.-J. Marigo, “Numerical experiments in revisited Griffith’s theory of brittle fracture,” *Journal of the Mechanics and Physics of Solids*, vol. 48, no. 4, pp. 797–826, 2000.

- [2] C. Miehe, M. Hofacker, and F. Welschinger, “A phase field model for rate-independent crack propagation: Robust algorithmic implementation based on operator splits,” *Computer Methods in Applied Mechanics and Engineering*, vol. 199, no. 45-48, pp. 2765–2778, 2010.
- [3] M. Ambati, T. Gerasimov, and L. De Lorenzis, “A review on phase-field models of brittle fracture and a new addition to the family,” *Computational Mechanics*, vol. 55, no. 2, pp. 383–405, 2015.
- [4] C. Pandey and A. Kumar, “Adaptive mesh refinement for phase-field fracture modeling using native Abaqus capabilities,” *Engineering Fracture Mechanics*, vol. 315, p. 110789, 2025.
- [5] C. Miehe, F. Welschinger, and M. Hofacker, “Thermodynamically consistent phase-field models of fracture: Variational formulation and numerical analysis,” *International Journal for Numerical Methods in Engineering*, vol. 83, no. 10, pp. 1273–1311, 2010.
- [6] O. C. Zienkiewicz and J. Z. Zhu, “A simple error estimator and adaptive procedure for practical engineering analysis,” *International Journal for Numerical Methods in Engineering*, vol. 24, no. 2, pp. 337–357, 1987.

A Variable and Symbol Table

Symbol	Description	Units
E	Young’s Modulus	kN/mm ² (GPa)
ν	Poisson’s Ratio	dimensionless
G_c	Critical Energy Release Rate	kN/mm (J/m ²)
l_0	Phase-field Regularization Length Scale	mm (μ m)
k_{res}	Residual Stiffness Multiplier (10^{-7})	dimensionless
d	Phase-field Damage Variable	[0, 1]
$g(d)$	Energetic Degradation Function	dimensionless
$\mathcal{H}$	Maximum Tensile Strain Energy History Field	kN/mm ²
h_{eq}	Equivalent Finite Element Size ($\sqrt{\text{Area}}$)	mm
K_0	Initial Structural Stiffness	kN/mm
F_{max}	Peak Reaction Force	kN
W_{trap}	External Trapezoidal Boundary Work	mJ (kN · mm)
θ_{kink}	Initial Crack Kink Angle (relative to horizontal)	degrees (°)
θ_{chord}	Overall Crack Chord Angle	degrees (°)

B Full PBS Job Execution Ledger

PBS Job ID	Job Name	Exit	Inc	Permanent Classification
1408540.mmaster02	M2_H1_CORR_DC	0	—	H1_DATACHECK_QUALIFIED
1408541.mmaster02	M2_H1_CORR_PROD	0	32	INVALID_ABI_DIAGNOSTIC
1408542.mmaster02	M2_H1_CORR_DC	0	—	H1_ABI_CORR_DC_QUALIFIED
1408543.mmaster02	M2_H1_CORR_PROD	0	144	H1_REFERENCE_PROD_QUALIFIED
1408548.mmaster02	M2_H2_CORR_DC	0	—	INVALID_PHASE_DOF_DIAGNOSTIC
1408549.mmaster02	M2_H2_CORR_DC	0	—	H2_DATACHECK_QUALIFIED
1408550.mmaster02	M2_H2_CORR_PROD	1	105	H2_SPATIAL_SENSITIVITY_QUALIFIED
1408551.mmaster02	M2_COARSE_PRE	0	10	MODE2_COARSE_PRE_QUALIFIED
1408554.mmaster02	M2_ADAPT_DC	0	—	MODE2_ADAPT_DC_QUALIFIED
1408555.mmaster02	M2_ADAPT_PROD	0	358	MODE2_ADAPT_PROD_COMPLETED
1408558.mmaster02	M2_TRI_PATCH_DC	0	—	TRI_PATCH_DC_QUALIFIED
1408559.mmaster02	M2_TRI_PATCH_PROD	0	2	TRI_PATCH_PROD_QUALIFIED

C Triangle Patch Verification Summary

The triangular patch test confirmed constant-strain mechanical parity with standard **CPE3** elements to $\leq 3 \times 10^{-9}$ kN, verified uniform degradation scaling $g(0.50) = 0.250000$, and confirmed identical internal energy tracking in **SDV17** (E_{frac}) and **SDV18** (E_{elas}).

D Publication Digitization Record

Data from Pandey & Kumar (2025) Fig. 13(a) was digitized into `pandey_kumar_2025_fig13a_digitized.csv` (SHA-256 0562087b...) and classified as `DIGITIZED_APPROXIMATE`.